TENDRARA PROJECT
GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS

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1. INTRODUCTION

This philosophy provides the basis for designing the flare, relief and blowdown systems for the Tendrara LNG Plant.

Four (4) flares are forecasted for the same burning pit:

1. HOT FLARE collecting relief and blow down header. Effluents are coming from gas process area, from inlet (SDV 2001) to dehydration outlet (SDV 3560).
2. FLARE collecting ammonia (NH_3) only
3. COLD FLARE collecting relief and vents from COLD STREAMS. Effluents are coming from liquefaction area, cold box inlet (SDV 3560) to LNG separator outlet (SDV 2201/2202)
4. HOT FLARE collecting CLIENT'S blow down from wells area, crossing the field through the same path of plant flares and ending to the same burning pit. This flare is estimated to be 10" diameter and it will be installed on pipe rack inside the Tendrara facility

The burning pit will be realized with dike, and located at safety distance from plant facility. Proper fence will segregate the area to avoid human access.

The burning pit is realised on South direction because of prevailing wind direction blowing from North.

Hot flares will have flanged tip and ignition/continuous pilots with thermocouples for status remote supervision.

The following system are collected to the Hot and Cold flares:

- Overpressure system to protect equipment against mechanical collapse.
- Blowdown system which is dedicated to equipment and piping depressurizing in order to reduce the hydrocarbon inventory and pressure within the process equipment in case of gas or fire detection.

Blow down and safety relief study will determine the loads discharged to the hot flare.

This document defines the following:

- basis for determining relief and blowdown flaring load
- the basis for blowdown and relief operations due to emergency conditions

No provision is made to discharge liquids in the event of an emergency shutdown (ESD). Usually flammable liquids are contained in the vessel during fire condition. Liquid inventory allows to slow down the temperature rise of the vessel exposed to a fire.

2. REFERENCE DOCUMENTS

ITF Document N.	TAM Doc. N.	Document Title
PRO03-ING-FR01	--	Project Design Basis
PRO10-ING-PR01/02/03/04	21044-PR-PFD-00001	Process Flow Diagrams
PRO10-ING-PR005/06/07/08/09/10/11	21044-PR-UFD-00001	Utility Flow Diagrams
PRO02-ING-FR01	21044-PR-HMB-00001	Heat and Materials Balance
PRO13-ING-PR01	21044-PR-PID-20001	Piping and instrumentation diagram Inlet feed raw gas
PRO13-ING-PR02	21044-PR-PID-33001	Piping and instrumentation diagrams Sweetening unit
PRO13-ING-PR03	21044-PR-PID-33002	
PRO13-ING-PR04	21044-PR-PID-33003	
PRO13-ING-PR05	21044-PR-PID-33004	
PRO13-ING-PR06	21044-PR-PID-33005	
PRO13-ING-PR07	21044-PR-PID-33006	
PRO13-ING-PR08	21044-PR-PID-31001	Piping and instrumentation diagrams Dehydration unit
PRO13-ING-PR09	21044-PR-PID-31002	
PRO13-ING-PR10	21044-PR-PID-31003	
PRO13-ING-PR11	21044-PR-PID-35011	Piping and instrumentation diagram Cold Box
PRO13-ING-PR12	21044-PR-PID-35012	Piping and instrumentation diagram Nitrogen Rejection Unit
PRO13-ING-PR13	21044-PR-PID-35001	Piping and instrumentation diagram MR compressor package PK-351-1
PRO13-ING-PR14	21044-PR-PID-35002	Piping and instrumentation diagram MR compressor package PK-351-2
PRO13-ING-PR15	21044-PR-PID-35003	Piping and instrumentation diagram MR compressor package PK-351-3
PRO13-ING-PR16	21044-PR-PID-35004	Piping and instrumentation diagram MR compressor package PK-351-4
PRO13-ING-PR17	21044-PR-PID-35005	Piping and instrumentation diagram MR compressor package PK-351-5
PRO13-ING-PR18	21044-PR-PID-35006	Piping and instrumentation diagram MR compressor package PK-351-6
PRO13-ING-PR19	21044-PR-PID-35007	Piping and instrumentation diagram MR makeup
PRO13-ING-PR20	21044-PR-PID-35008	Piping and instrumentation diagram KO drum, compressors (simplified)
PRO13-ING-PR21	21044-PR-PID-35009	Piping and instrumentation diagram ORS

PRO13-ING-PR22	21044-PR-PID-35010	Piping and instrumentation diagram Aircooled MR cooler, ammonia MR cooler, MR separator, MR pumps
PRO13-ING-PR23	21044-PR-PID-22001	Piping and instrumentation diagram LNG unloading skids
PRO13-ING-PR24	21044-PR-PID-22002	
PRO13-ING-PR25	21044-PR-PID-22003	Piping and instrumentation diagram LNG separator
PRO13-ING-PR26	21044-PR-PID-22004	Piping and instrumentation diagram LNG tank
PRO13-ING-PR27	21044-PR-PID-37001	Piping and instrumentation diagram Vaporizer skids
PRO13-ING-PR28	21044-PR-PID-41001	Piping and instrumentation diagram Hot oil unit
PRO13-ING-PR29	21044-PR-PID-36001	Piping and instrumentation diagram BOG compressor
PRO13-ING-PR30	21044-PR-PID-36002	Piping and instrumentation diagram BOG compressor
PRO13-ING-PR31	21044-PR-PID-36003	Piping and instrumentation diagram BOG compressor
PRO13-ING-PR32	21044-PR-PID-36004	Piping and instrumentation diagram BOG recycle compressor
PRO13-ING-PR33	21044-PR-PID-36005	Piping and instrumentation diagram BOG recycle compressor
PRO13-ING-PR34	21044-PR-PID-34001	Piping and instrumentation diagram Ammonia compressor package PK-341-1
PRO13-ING-PR35	21044-PR-PID-34002	Piping and instrumentation diagram Ammonia compressor package PK-341-2
PRO13-ING-PR36	21044-PR-PID-34003	Piping and instrumentation diagram Ammonia compressor package PK-341-3
PRO13-ING-PR37	21044-PR-PID-34004	Piping and instrumentation diagram Ammonia compressor package PK-341-4
PRO13-ING-PR38	21044-PR-PID-34005	Piping and instrumentation diagram Ammonia condensers, receiver, compressor simplified
PRO13-ING-PR39	21044-PR-PID-50001	Piping and instrumentation diagram Instrument air compressor / N2 generator
PRO13-ING-PR40	21044-PR-PID-41001	Piping and instrumentation diagram Hot oil unit buffer
PRO13-ING-PR41	21044-PR-PID-42001	Piping and instrumentation diagram Fuel gas conditioning
PRO13-ING-PR42	21044-PR-PID-43001	Piping and instrumentation diagram Diesel storage and distribution
PRO13-ING-PR43	21044-PR-PID-23001	Piping and instrumentation diagram Flare
PRO13-ING-PR44	21044-PR-PID-23002	
PRO13-ING-PR45	21044-PR-PID-23003	

PRO13-ING-PR46	21044-PR-PID-55001	Piping and instrumentation diagram Closed drain
PRO13-ING-PR47	21044-PR-PID-53001	Piping and instrumentation diagram Oily water
PRO13-ING-PR48	21044-PR-PID-54001	Piping and instrumentation diagram Raw water
PRO02-ING-FR02	21044-PR-POD-00001	Process Description

Table 1. Reference documents.

3. INTERNATIONAL CODES AND STANDARDS

API Codes

Code No.	Title	Edition
Std 520	Sizing, selection and installation of pressure relieving devices in refineries - Parts I and II	2020
Std 521	Pressure-relieving and depressuring systems	2020

Table 2. API Codes.

OTHER STANDARDS

Code No.	Title	Edition
ASHRAE 15	Safety standards for refrigeration systems	2019
EN 13136	Refrigerating systems and heat pumps - pressure relief devices and their associated piping - methods for calculation	2020-08
NFPA59A	Standard for the production, storage and handling of liquefied natural gas (lng)	2019

Table 3. Other applicable standards.

4. PRESSURE RELIEVING SYSTEMS

Relief systems allow the discharge of fluid inventory to maintain the contents within safe design pressure limits. They are typically used to protect the equipment against operation upsets such as blocked outlet, gas blow-by and fire scenarios. The relief system will be designed to dispose the relieving flow in a safe and reliable manner. Relief of process fluids is primarily intended to provide protection against the mechanical failure of the equipment and piping due to over pressurization.

4.1. EVALUATION OF RELIEF REQUIREMENTS

For each safety/relief valve, the maximum relieving requirement has been determined for each of the applicable causes described in API 521, Section 4. Relieving scenarios which only occur as a result of double jeopardy should not be considered if the two events are truly independent. Common mode failures have also been considered.

The data for all pertinent relief conditions have been stated on the safety/relief valve data sheet.

In general, the initial protection level previewed for this project is a pre-alarm PAH displayed on the Process Control System to alert operators about an abnormal increase of pressure. The following protection level previewed for this project is represented by an alarm PAHH which will generate actions (Process Shut-Down, PSD) to protect against a further and uncontrolled increase of pressure.

A third protection level is represented by a mechanical relief device (PSV).

The following conditions of overpressure are applied for relief valves sizing:

- External Fire

  On process vessels in the blocked condition. If the vessel has some liquid inventory, liquid inside will vaporize and latent heat absorb the heat through the vessel wall when it is exposed to an external fire. If the liquid inventory is low or if the content is a gas, the heat from external fire is not dissipated by vaporizing liquid.

  For fire case in the Amine regenerator system, we consider that the PSV on the column can relieve the vapor generated in the column, reboiler, condenser and amine reflux accumulator.

- Blocked Outlets

  The closure of a block valve on the gas outlet line of a pressure vessel can cause the vessel's internal pressure to rise and exceed its maximum design pressure (MAWP).

  Blocked outlet condition of liquids outlets on two/three phases separator will lead to equipment shut down by LSHH.

  For the case of blocked liquid outlet, the liquid level in the pressure vessel may rise. If the surge time for liquid overflow from the vessel is less than 15-20 minutes, then a blocked liquid outlet is a valid over-pressure scenario. If the surge time is longer, it can be assumed that operators have sufficient time to take action to avoid an overflow.

- Gas blow by

This condition will happen when liquid level in vessel is/ could be lower than minimum value (LALL) and gas is entrained in the liquid line. For all critical loops, two levels of protection shall be provided, a dedicated SDV actuated by LALL (LSD logic) and a PSV sized for this condition.

4.2. COMMON SOURCES OF OVERPRESSURE FOR RELIEF INTERVENTION

Below a summary of all the possible general causes of overpressure adopted for relief system calculations:

- *External Fire*

Fire is a common cause of over-pressure in pipes and vessels. Analyzing for fire case assumes that the vessel exposed to fire is blocked in condition.
If the vessel has some liquid inventory, liquid inside will absorb the heat through the vessel wall when it is exposed to an external fire. If the liquid inventory is low or if the content is a gas, the heat from external fire is not dissipated by vaporizing liquid.
For fire case in the Amine regenerator system, we consider that the PSV on the column can relieve the vapor generated in the column, reboiler, condenser and amine reflux accumulator.

- *Blocked Outlets*

The closure of a valve on the gas outlet line of a pressure vessel can cause the vessel's internal pressure to rise and exceed its maximum design pressure (MAWP).

- *Utility Failure*

Utility failures such as loss of power or cooling water can affect all equipment using power or cooling water at the same time. To be safe, the flare header should be designed and sized based on the maximum relief load that could result by a potential utility failure.

- *Electric Power Failure*

For the Tendrara plant, 7 main generators (gas engine driven) are planned for a total electrical power of approximately 14 MW. In addition, 2 diesel generators (1 MW each) are provided for start-up in the absence of gas or the lack of 3 gas generators.

In the event of a lack of fuel gas, the system is stopped and therefore the diesel units will guarantee power supply to emergency users.

In the event of a malfunction of 3 gas engine driven generators, the diesel generators can guarantee the continuation of operations without interruption of power.

In the event of a short circuit on the main bars, there will be no loss of power but only the stop of the process (process shut down).

General power failure affects the entire process plant and the immediate consequences are as follows:

- Electrical motor-driven pumps and compressors will stop. Thus, feed input, product output and reflux flows will stop. Instrument air will be interrupted and network pressure will drop causing all the pneumatic devices to switch into « fail condition », ''fail close'' or ''fail open''.
- Motor-driven fans for air coolers, cooling heat exchangers, process refrigeration units and burners will stop.
- Motor-operated or solenoid valve will fail in closed position.
- Diesel engine driven power generators starts to feed the emergency power users, such as UPS and all others headed under the emergency power generation load list.

- *Instrument Air failure*

Instrument air supply is provided with an air reserve. This, and the high air pressure may, after a power failure, be able to maintain an adequate air supply for a short period of time. Normally the air receiver should be able to maintain 10 - 15 minutes of air supply after the air compressor has tripped for power failure.

- *Fuel Gas / condensate*

In the facility Fuel gas is taken from different sources to assure that the required flow is always available for supply to the users.

TOTAL fuel gas sources comprise:

F105 + F8 + F101 + F103

(Please refer to Process Flow Diagrams PRO10-ING-PR01/02/03/04 for stream identification)

Main headers detail:

- F105: coming from E-371, collecting F86 (from Inlet raw gas separator V-201, used only for start up) + F36 (from Dehydration unit outlet) + DC1 (from NGL separators in Cold Box CB-356).

- F8: coming from V-312 Regeneration wet gas separator.
- F101: coming from PK-361-C1A/B BOG compressor.
- F103: coming from Nitrogen Rejection Unit T-357.

Overtemperature condition (TSHH) as well as fuel supply low pressure are managed through the power generators control system and burner management system (BMS) for heating equipment.

- *Abnormal Heat Input*

Some possibilities include:

- The supply of heating medium, such as fuel oil or fuel gas to a fired heater, can be switched from medium heating value to higher heating value.
- The control valve for the fuel supply fails fully open or fuel control valve leakage.
- Abnormal Vapor Input (Gas blow by).

This is especially important if the upstream equipment is under high pressure. The system may not be able to discharge the extra vapor from the upstream process and eventually become over-pressured.

- *Thermal Expansion*

It is rare that a vapour-filled line or vessel would be over-pressured by external heat source, other than fire. On the other hand, when a liquid is blocked in a vessel or line pipe, external heat input can cause liquid temperature, and hence volume, to rise.

Common causes are:

- The liquid being blocked in has a vapour pressure higher than the relief valve set pressure, thermal expansion is a potential cause of over-pressure. On all cryogenic lines and equipment thermal expansion due to heat input from the environment is possible in case of cryogenic liquid trapped in, thermal relief valves are installed to prevent this possibility.
- The heat exchanger cold side is filled due to hot side is still flowing.
- A pipeline or vessel is filled with liquid at ambient temperature and is heated by direct solar radiation. Even if the liquid being blocked in has a vapor pressure lower than the relief valve set pressure, thermal vaporization is a potential cause of over-pressure.

- *Accumulation of Non-Condensable*

During normal operations, non-condensable do not accumulate in the system, because they are discharged along with the process streams or through a vent. Accumulation may occur under the following conditions:

- The normal non-condensable vent is blocked.
- The piping configuration or equipment has a pocket in which non-condensable accumulates.

The accumulated non-condensable can blanket a condenser and result in loss of cooling duty.

Refrigeration loops, UNIT 340 and 350 may accumulate non condensable if the first filling of refrigerant is not done properly. Refrigeration vendor standard procedure will be followed onsite to avoid non-condensable to be trapped in the system (vacuum procedure as per refrigeration standard will be followed on site), provision for non-condensable venting is included in the design. The refrigeration loops (UNIT 340 and 350) run with operating pressure above atmospheric so once a proper refrigerant filling and non-condensable evacuation are completed, we don't foresee this event to be possible and/or frequent.

- *Valve Malfunction*

Only one valve malfunction needs to be considered at a time, if other valve malfunctions are unrelated. Examples of valve malfunction include:

- Check valve backflow - this is a rare case but should be considered when a high pressure differential is present between the inlet and outlet of a check valve, a pump or other piece of equipment.
- Inadvertent valve operation (open/close, bypass) - a valve may be wrongly operated to a position that is opposite to its intended normal operating position. This is a very common over-pressure scenario, and is the result of human error. Very often, a fully open bypass valve can pass a greater fluid volume than the main control valve.
- Control valve fails fully open or closed - caused by mechanical or electronic signal failures.

API 520 and 521 do allow one to take credit for use of "car seals open" (CSO) valves. Such valves maintain an open position all the time. The reverse is the "car seals close" (CSC) valves that are in close position all the time.

- *Upstream Relieving*

The following scenarios should be evaluated:

- When an upstream vessel is relieving by discharging fluid into a downstream vessel, the downstream vessel must be designed to handle the pressure and volume of the incoming stream from the upstream vessel without over-pressuring itself. If the upstream vessel does not have adequate relief capacity, the downstream vessel needs to have a pressure relief valve of its own.

• 2 vessels are connected by an open path. The first has its own pressure relief valve and it discharges into a flare header. The second will experience the impact from the relieving pressure of the first vessel relieving.

- *Loss of aircooler*

  For the column with condenser and accumulator it will be evaluated that the aircooler stops and vapor has to be relieved through column PSV.

- *Loss of refrigeration outside the cold box and Atmospheric LNG tank pressurization*

  In case of loss of refrigeration inside the cold box it is possible that high pressure vapor may be sent to the atmospheric LNG tank and pressure the LNG tank. This condition is considered, and the following precautions are implemented: monitoring of the high-pressure LNG temperature, first LNG flash in a Nitrogen Rejection Unit. The level in the Nitrogen Rejection Unit act as a hydraulic seal against LNG atmospheric tank pressurization. The LNG from the NRU is further expanded in a LNG separator (V-221), also this separator act as hydraulic seal against LNG atmospheric tank pressurization. The atmospheric LNG tank have 2x100% pilot operated PSV with proportional action designed in all possible vapor pressure release scenario as required by NFPA 59A.

- *Heat exchanger tube rupture*

  Complete tube rupture, in which a large quantity of high-pressure fluid flows to the lower-pressure heat exchanger side, is a remote but possible contingency. Minor leakage can seldom overpressure a heat exchanger during operation; however, such leakage occurring where the low-pressure side is closed in can result in overpressure and should be assessed.

4.3. Type of Relieving Device

Depending on the over pressure protection requirements, the following three types of pressure relief devices may be used.

Wherever possible, consistent with the desire to minimize the size of flare pipework, the maximum relief valve backpressure should be limited to 10% of the PSV's set pressure so as to maximize the use of conventional PSVs. Balanced bellows PSVs will be used for cases in which the PSV backpressure exceeds 10% of its set pressure. Pilot operated valves may also be considered.

1) Direct Actuated Relief Valves

The direct actuated spring-loaded relief valve is the preferred type because of its durability and mechanical simplicity.

For applications where a high back pressure is expected (more than 10% of set pressure) balanced bellows type of relief valves should be used. Balanced bellows RVs typically may be used up to a maximum allowable back pressure of approx. 50% of set pressure, the exact maximum depending on the valve model.

2) Pilot Operated Relief Valves (PORV)

Pilot operated relief valves are most suited to clean gas service and should only be used if a direct actuated relief valve is unsuitable, such as when the flare system back pressure is excessive or if there is a reduced margin between maximum operating and design pressure.

PORVs have a number of advantages over direct actuated relief valves:

- Set pressure is unaffected by back pressure.
- If a remote sensing line is used PORVs can accept higher inlet losses than direct actuated relief valves without chattering.
- Tight seating and hence lower leakage (valve will not lift until set pressure is reached).
- Valve sizes and weights less than equivalent capacity conventional valves.

If PORVs are used, special care must be taken to ensure that there is no possibility of blockage of the pilot or sensing line with hydrates, ice, wax or solids. Only non-flowing pilot types are acceptable

“Backflow preventers” should be used if the maximum expected flare system pressure can exceed the normal operating pressure upstream of the valve and reverse flow through the relief valve would be unacceptable.

3) Bursting Discs

Bursting discs shall not normally be installed since they cannot be tested without removal and flow continues until all pressure is relieved. They should only be used:

- when relief valves can not provide an adequate response
- to protect the inlets of safety/relief valves from continuous contact with corrosive or solidifying process fluids.

4.4. Relief Valve Set Pressure and Max Allowable Accumulated Pressure

The Set pressure is the pressure at which the relief device begins to activate; usually it is set at the MAWP of the vessel protected.

Multiple relief valves may have staggered set pressures to avoid damage due to chatter. Usually the first safety valve is set at 100% of the MAWP and it is smaller than additional valves so to minimize the product loss.

Note however, that the pressure loss in the RV inlet system, added to the relief valve inlet pressure at relieving capacity, should not exceed the Maximum Allowable Accumulated Pressure (MAAP) of the protected equipment.

In *non-fire contingencies* the accumulated pressure may be limited to 110% of the MAWP in vessels that are protected by only one valve. If the MAWP lies between 15 and 30 psig (1 to 2 barg), the allowable accumulation is fixed to 3 psi (0.2bar).

In vessels protected by more valves, the accumulated pressure may be limited to 116% of MAWP or to 4 psi (0.27bar) if the MAWP lies between 15 and 30 psig (1 to 2 barg).

In *fire contingencies* the accumulated pressure may be limited to 120% of the MAWP in vessels that are protected by only one valve. Safety valves sized for the fire case may be used in non-fire situation, provided that they satisfy the constrain on the accumulated pressure of 110% (one valve) and 116% (more valves) respectively.

4.5. RELIEF VALVE SIZING AND SELECTION

The required relief valve “effective” area should be based on the maximum relief requirement (see Section 2.1) and determined using the sizing methods given in API 520 Part I, ASHRAE 15 (applicable to refrigeration system) and/or EN13136 (applicable to refrigeration system).

The required “effective” area should be used to select one or more relief valve(s) to meet this requirement without providing an unnecessarily large relief margin to avoid damage due to chatter.

The sizing of the flare system will account for the maximum allowable back pressure on pressure relief valves. Normally taken as 30% of set pressure for balanced bellows type valves and 10% for conventional valves.

4.6. RELIEF VALVE ISOLATION PHILOSOPHY

Where duty / standby relief valves are installed then full-bore isolation valves will be provided in the inlet to, and outlet from, each relief valve for maintenance purposes.

In order to ensure that sufficient relief valves are in service to meet the overpressure protection requirements, the duty/standby relief valve isolation valves shall be locked in position as follows:

Relief Valve Installation	Duty PSV	Standby PSV
Duty/Standby		
inlet isolation valve SDV	LO	LC
outlet isolation valve SDV	LO	LO
Single PSV		
inlet isolation valve	Usually not provided (LO when provided)	-
outlet isolation valve	LO	-

Table 4. Relief valve isolation.

For refrigeration and cryogenic service a changeover (COV) valve can be used upstream the PSVs, the changeover valve doesn't need to be Lock open (LO) because it guarantees always at least one way fully open.

4.7. Relief Valve Bypass

Each relief valve installation should be provided with a 1" bypass to enable the upstream equipment to be manually depressurized. The bypass should include a locked closed all valve followed by a gate/globe valve.

4.8. Relief Valve Location

Relief Valves shall be located such that the contents of the protected system do not interfere with relief valve operation. This is particularly important for streams containing solids or high pour point liquids. Relief valves shall be located above the protected system. Relief valves shall be located above any discharge header such that drainage of any liquids is into the header system. There shall be no pockets in the relief piping.

The pressure relief valve should be located as close to the source of pressure as practicable, to limit inlet pressure losses and hence prevent valve chatter. API 520 recommends that inlet losses be restricted to 3% of set pressure. If this limit cannot be met, consideration should be given to using pilot operated RVs with remote sensing.

If a vessel is fitted with a demister or any other outlet device which may be susceptible to blockage, the relief connection shall, where practical, be located upstream of the device.

4.9. Relief Valve Configuration

Different relief valve configurations may be used depending on the equipment on-stream factor and testing interval required for the relief valve.

Whether a shutdown is not practicable or would lead to an unacceptable loss of production, the design includes the isolation valves and n°2 PSVs (duty/stand-by set-up). Isolation valves will be Locked open or closed according to the description in 4.6.   and a procedure shall be placed to make sure that the status of the isolation valves is continuously monitored, in order to ensure that one of the PSVs is in service at all times. In case of using a changeover valve instead of the inlet isolating valves, the changeover valve itself always assures that one of the two ways is open. Downstream of the PSVs, both outlet valves shall be LO.

4.10. RELIEF VALVE DEVICE SPARING

Spare relief valves shall be installed in locations where it is unacceptable for an item of equipment or system to be shut down before the relief valve can be maintained. Therefore, spare relief valve is not required when 100% spare equipment is provided or for minor thermal relief and utility duties, but only if this will cause no or minimal disruption to normal plant operation. The use of bursting discs is not preferred.

5. BLOWDOWN SYSTEMS

5.1. GENERAL

Where fire is controlling, it is appropriate to provide depressurizing on all equipment that processes light hydrocarbons and set the depressurized rate to achieve 6.9 barg or 50% of the design pressure, which ever is lower, in 15 minutes.

UNIT 340 contains ammonia as refrigerant and for this system we are not providing any depressurizing system different than PSV in compliance with ASHRAE 15 and EN 378 (standard of refrigerating system). Any refrigeration system contains a boiling fluid that generates autorefrigeration effect during a fire event.

UNIT 350, Mix Refrigerant loop also does not have any additional depressurization system different that PSV's for the same reason.

Depressurizing systems are principally required to reduce the risk of loss of equipment integrity during a fire or to reduce a local loss of containment arising from a leak which may otherwise lead to escalation and the risk of catastrophic structural failure.

A pressure relief valve cannot provide depressurizing; it will merely limit the pressure rise to a given value under emergency conditions. In evaluating a vapor depressurizing system for fire load, it is particularly worthwhile to consider the possibility of a fire occurring around a vessel that contains both liquid and vapor. The unwetted portion of the vessel will probably reach a temperature at which the strength of the material will be reduced. In this instance, the pressure relief valve will not protect against rupture; whereas, a vapor depressurizing system could reduce the pressure to a safe level.

Depressurization also reduces the hydrocarbon inventory which would otherwise be released should the system subsequently fail.

Blowdown systems work in conjunction with the shutdown system to isolate and de-pressurize process equipment in the event of a fire or incident where the process inventory is required to be removed, e.g. fire or major gas leak. In the case of a fire, this is required to reduce the equipment wall stress to a level sufficient to prevent failure at the high temperatures associated with exposure to external fire.

Blowdown facilities will not be provided on hydrocarbon gas systems with normal operating pressure below 6.9 barg, nor on liquid filled systems, nor on LPG systems.

Blowdown from vessels should be positioned to minimize reverse-flow and ensure that blowdown rates cannot damage vessel internals.

Emergency blowdown (BDV) during a fire is necessary to minimize the risk that the pressure within the system exceeds the rupture pressure at any point of the system at any time during blowdown, allowing for the reduction of yield strength with increasing temperature. Equipment particularly at risk includes pressurized equipment with thin walls (especially piping), equipment with a low liquid inventory and equipment without fire proof insulation.

Liquid hold-up required for normal operations can be effective in keeping the vessel wall cool and does not necessarily require drainage systems for its removal. Hence, the normal amount of liquid will be retained in order to slow down the temperature rise of the system exposed to fire and hence to delay the reduction in material strength, so no liquid blowdown is applied in the project.

5.2. PIPELINE DEPRESSURIZATION (BY OTHERS)

NOT USED.

5.3. ACTIVATION

Emergency depressurization of each fire zone or of the whole facility (small installation) can be executed manually or automatically after the emergency shutdown sequence (ESD) has been activated automatically:

- by fire detection system or
- by smoke detection system

or manually by operators:

- by the pushbutton on the Safety Panel
- by one of the several pushbuttons of the E.S.D. circuit along the perimeter of the installation.

Blowdown valves (BDVs) are considered to be emergency valves, therefore must fail open.

All depressurizing valves (BDVs) within the fire zone or the installation will be de-energized simultaneously by activating manually/automatically the BD system.

Each blowdown device will comprise of a fail open pneumatic valve with a downstream restriction orifice sized to allow the blowdown rate required by the depressurizing criteria. The blowdown flow is discharged into the appropriate flare collection system, the same of the Overpressure and Safety valves discharge.

5.4. DEPRESSURIZING RATES

The depressurizing orifice must dispose of vapor due to:

1) initial vapor inventory
2) liquid flashing due to the reduction in pressure
3) vapor generated from liquid by heat input from a fire

Pressure rating changes downstream of the blowdown valve. Piping from the actuated valve downstream will be suitable for cold temperatures if required. Checks will be made on systems with high CO_2 content that solid CO_2 will not form during depressurization.

The selection of minimum equipment design temperatures will be verified as part of the overall blowdown study. The impact of depressurizing a static system that has cooled to minimum ambient conditions will be assessed if appropriate.

To determine the vapor quantities contributed by the factors mentioned above, it is necessary to establish a liquid inventory and a vapor volume of the system. Subject to corrections made necessary by the final plant design, the following initial assumptions should be used:

1) Initial pressure equal to maximum operating pressure (taken to be at High Pressure Trip or, for compressors, initial settle out pressure).

2) To determine the vapor depressurizing flowrates, it is necessary to establish a liquid inventory and the vapor volume of the system. This shall include all the facilities located in the fire zone and all equipment outside the fire area which, under operating conditions, are in open connection with the facilities located within the fire area.

3) The liquid inventory of separators, flash drums, knock-out vessels etc., shall be based on normal operating levels.

4) The liquid inventory of columns can be estimated as the normal (NLL) column bottom and draw-off tray capacity plus the normal tray liquid hold-up (6cm)

5) The effective fire heat input surface area of a piece of equipment exposed is the area wetted by the internal contents of the equipment within a given fire zone.

6) The sizing of depressurizing systems shall be based on the assumption that during a fire, all feed and output streams to and from the equipment are stopped and all internal heat sources within the process have ceased.

The peak load will occur at the beginning of the event. Heat input from fire should be considered only for equipment in the most significant fire zone. It is recognized that during a fire incident it is unlikely that the fire will simultaneously impinge on all vessels/equipment within a manageable area.

The rate of depressurizing relevant equipment will be as recommended per API 521.

5.5. Evaluation of Depressurizing Rate

The method used to evaluate the depressurizing rate is detailed in API 521.

The blowdown calculation will normally be carried out in HYSYS to determine equipment pressures and initial and intermediate blowdown rates and will take into account the following:

- hot blowdown, when the system to depressurize is subjected to fire;
- cold blowdown, when the system blowdown commences from minimum operating temperature (if lower) in adiabatic case. The impact of depressurizing a static system that has cooled to minimum ambient conditions will be assessed if appropriate.

Blowdown initial conditions will be chosen to ensure a conservative design. The blowdown analysis will consider vaporization of any associated liquid due to the reduction in pressure. Heat input from fire should be considered only for equipment in the most significant fire zone.

5.6. Depressurizing Device/Sizing

The depressurizing device is a restriction orifice associated with an upstream actuated ball valve. Valves used normally as pressure control valves will not be used as blowdown valves. Depressurizing devices will be sized using the maximum anticipated back pressure profile.

5.7. Depressurizing Line Sizing

The Mach Number at the outlet of a blowdown device should not exceed 0,7 inside the tail pipe; at flare header it is accepted a maximum velocity of 0,5 Mach because the depressurization represents a peak, short term, infrequent flow.

5.8. Depressurizing Device Location

The location of depressurizing valves shall be governed by the same considerations as relief valves and they may discharge into the same disposal system (PSV header) as the relief valves on the equipment under consideration.

Depressurizing devices shall generally be located above protected equipment to avoid liquid entrainment. Particular attention should also be paid to the position of non-returns valves when locating depressurizing valves, to ensure that equipment downstream of the non-return valve cannot be isolated from the depressurizing valve. Depressurizing valve systems also require periodic testing, including operation and maintenance and should therefore be easily accessible.

5.9. Depressurizing Valve Failure Mode

Emergency blowdown valves shall fail to the open position. Specification of all emergency blowdown valves to fail to the open position gives the potential for all such valves to fail safe simultaneously e.g. upon loss of instrument air supply or UPS.

6. DESIGN OF PIPING UPSTREAM OF RELIEF/BLOWDOWN DEVICE

6.1. General

Piping upstream of a relief/blowdown device will be designed with as few restrictions as possible and will be self-draining.

6.2. Depressurizing Valves

The pressure drop between the protected system and the depressurizing device is important but not as critical as for a relief valve since chattering is not a problem. However, excessive pressure drop between protected system and device will adversely affect the capacity of the device and should be avoided. Allowance shall be made in sizing the depressurizing device for upstream pressure losses. The depressurizing valves will normally be actuated from a remote safe nearby location as well as the control room. In general, all inlet and outlet piping to and from the depressurizing valves shall comply with API 520/521.

The location of the depressurizing valves shall be governed by the same considerations as relief valves and they may discharge into the same disposal system as the relief valves on the equipment if relieving temperature permits.

7. DESIGN OF PIPING DOWNSTREAM OF RELIEF/BLOWDOWN DEVICE

7.1. General

Piping downstream of a relief/blowdown device should be designed using the following criteria:

1) Fall towards the burning pit shall be 0.2%.
2) The piping should have a minimum of restrictions and should not be pocketed.
3) The backpressure on the RV/BDV shall not restrict the required capacity.
4) Piping system design pressure will not be exceeded.

7.2. Sizing of Downstream Piping System

The release of high-pressure gas to a lower pressure system is usually accompanied by a significant drop in temperature and possibly the formation of liquids. Then, the flare system shall be designed to be free draining and have no pockets.

The following criteria will be accounted for:

Headers:

1) Flare headers shall be continuously sloped towards the burning pit.
2) For an atmospheric vent collection system, a drain shall be provided at the lowest point. The drain shall be piped to a safe location and may contain a locked-open isolation valve in an easily accessible location.
3) The flowing velocity in flare header is accepted to be less than 0,7 to prevent fatigue due to acoustically induced vibration. It will be chosen a nominal maximum of 0,5 Mach.

Laterals:

4) Individual tailpipes for relief valves should be sized using the rated flow of the installed valve.
5) Flare header laterals (tail pipes) connect the relief valves to the flare header itself and, in accordance with API 520, the diameter shall not be less than the relief valve outlet size. This increase in line size must be immediately downstream of the relief valve outlet.
6) Normally, laterals should be “swept in” at 45º to the flare header in the vertical plane.
7) Where laterals are of a lower grade material than the main flare header due to temperature or service considerations, then the piping specification break of the lateral should be a minimum of 2 m from the flare header.
8) The flowing velocity in sub-headers and tail pipes will be normally less than 0,7 Mach.

Headers and Laterals:

1) The pressure loss shall be calculated using the rated capacity of the installed pressure relief valves for the tail pipes (Laterals) and the relieving flow rate (of the governing case) for the main flare header, flare KO Drum and flare stack.
2) Outlet pipework system back pressure calculations shall be performed following a comprehensive review of all potential flare/vent system loads. This shall cover all relief, blowdown and normal controlled flaring/venting scenarios and identify which loads can be coincident.

8. PROTECTION OF LNG ATMOSPHERIC TANK

8.1. General

The LNG atmospheric tank can suffer rupture in case of internal pressurization generated by an upset scenario or vacuum.

The LNG tank is provided with 2x100% PSV on the inner tank sized for all possible pressurization event:

1. Loss of refrigeration in cryogenic cold box;
2. LNG loading pump recirculation to the LNG tank;
3. Boil off gas compressor upset conditions (BOG trip scenario)

The LNG tank is provided with a pressure build up coil to avoid vacuum operation of the LNG tank, when LNG production unit and BOG compressors are out of service and unloading skids are in operation.